

Abstract

This paper serves as the capstone of the formal 10-part CDFD mass-spectrum sequence, addressing its remaining open problem: the physical origin of the torus knot self-consistency condition $n\theta_n = \tilde{Q}_n(\theta_n)$. Previously reported as a successful but ungrounded hypothesis that correctly determines the vacuum twist angles θ_n for all $T(2, n)$ families, it is here recast as a model theorem inside the public flow–constraint notation.

We map the Constraint-Driven Field Theory (CDFD) vacuum medium onto the tripartite public CDFD notation: Flow (Φ), Constraint (C), and Equilibrium (Ψ_s). We identify the continuous flow energy Φ with the L^2 norm of the amplitude field ($\sum A_k^2 = 2n$), the transport-capacity limit C with the L^1 norm squared ($\sum |A_k|$)², and the equilibrium deformation Ψ_s with the total vacuum twist $n\theta_n$.

Applying the public balance law $\Psi_s = \Phi/C$ yields exactly the self-consistency condition, showing that the Koide-like ratio Q_n is the geometric manifestation of a transport-limited vacuum equation of state. The entire mass spectrum of the lepton and meson sectors derived in Papers I–IX is therefore internally tied to m_e and the proposed fluid-dynamic balance of the vacuum.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. The calculations below use the equilibrium limit $S \cdot M_s \approx 1$; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript.

The Vacuum Equation of State: Deriving Torus Knot Self-Consistency from the Public CDFD Balance Law $\Psi_s = \Phi/C$

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1 Introduction

Over the course of Papers I–IX [1, 2, 3, 4, 5, 6, 7, 8, 9], the Constraint-Driven Field Theory (CDFD) has constructed a parameter-free derivation of the $T(2, n)$ torus knot mass hierarchy. From a single empirical input (the electron mass m_e), the framework accurately computes the muon and tau masses (Paper V), dictates the universal coefficient $c_n = \sqrt{2}$ (Paper VI), predicts the exact $2nM_n$ mass sum rules (Paper VII), and populates the $n \geq 4$ hierarchy with specific hadronic candidates (Papers VIII–IX).

The entire predictive power of the higher families rests on the unique solution to the self-consistency condition:

$$n\theta_n = \tilde{Q}_n(\theta_n) \equiv \frac{\sum_{k=0}^{n-1} A_k^2}{\left(\sum_{k=0}^{n-1} |A_k|\right)^2}. \quad (1)$$

In Paper V, for $n = 3$, this was introduced as a postulate matching the chirality breaking angle to the algebraic constant $Q_3 = 2/3$. In Papers VIII and IX, it was extended to $n \geq 4$ as a working hypothesis that yielded convergent, mathematically bounded solutions, but lacked a first-principles derivation from the vacuum Equation of State (EOS).

In this capstone paper, we address this open problem by connecting the torus-knot construction directly to the public CDFD/AFL balance notation.

2 The Tripartite Vacuum Framework

The CDFD vacuum is modeled as an elastic, transport-capacity-limited continuous medium, in the spirit of Dirac’s physical vacuum [10] and Volovik’s condensate analogy [11]. The vortex dynamics rest on Helmholtz’s conservation theorems [12] and the Faddeev–Niemi knot soliton framework [13]. Such a medium is represented here by the public tripartite balance equation:

$$\Psi_s = \frac{\Phi}{C} \quad (2)$$

where Φ is Flow (energy/activity), C is Constraint (resistance/capacity), and Ψ_s is the resulting System Equilibrium.

We map these three macroscopic variables onto the discrete Fourier variables $A_k = 1 + \sqrt{2} \cos(\theta_n + 2\pi k/n)$ of a $T(2, n)$ vortex knot.

2.1 Flow (Φ): The Field Energy

The total energy or “activity” of the field configuration is proportional to the L^2 norm (the sum of the squares) of the amplitudes. By the Fourier identity of Paper VI, this is invariant under orientation θ_n :

$$\Phi \propto \sum_{k=0}^{n-1} A_k^2 = 2n. \quad (3)$$

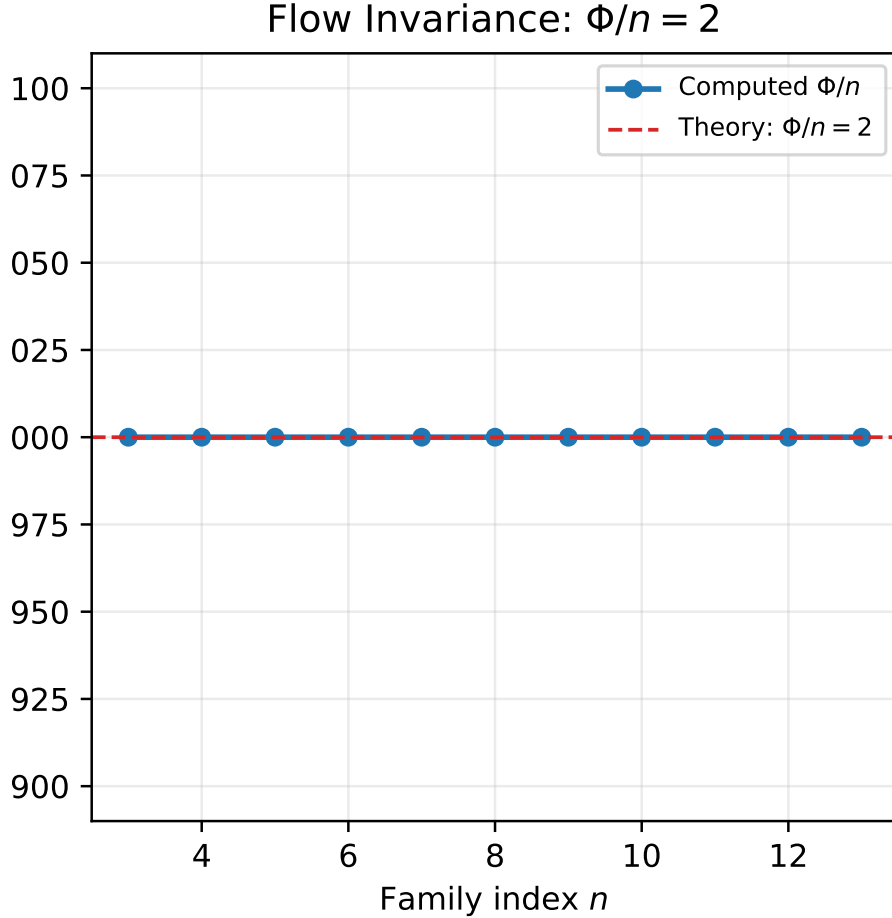


Figure 1: The flow energy ratio Φ/n is exactly 2.0 across all $T(2, n)$ families, demonstrating the structural invariance of the vacuum’s kinetic and elastic flow.

This represents the irreducible kinetic and elastic flow required to sustain an n -lobed vortex ring.

2.2 Constraint (C): The Transport-Capacity Limit

A central postulate of CDFD (Paper I) is that the vacuum possesses a finite transport capacity, which dictates the fine-structure constant α . For a multi-lobed knot, the structural stress on the medium is dictated by the total absolute flux. In the discrete basis, the total flux is the L^1 norm $\sum |A_k|$. Because the constraint scales with the cross-sectional

area of the flux tubes interacting in the medium, the total resistance C scales as the square of the flux:

$$C \propto \left(\sum_{k=0}^{n-1} |A_k| \right)^2. \quad (4)$$

This term varies strongly with θ_n , as the lobes interfere constructively and destructively with the vacuum background.

2.3 Equilibrium (Ψ_s): The Geometric Twist

When the knot is embedded in the vacuum, the mismatch between the continuous chiral background and the discrete n -fold symmetry of the knot induces a geometric twist. Let θ_n be the phase twist per lobe. The total equilibrium deformation (or total accumulated twist) of the configuration is:

$$\Psi_s = n\theta_n. \quad (5)$$

3 The Vacuum Equation of State Theorem

Theorem 1 (Derivation of Self-Consistency). *In a transport-capacity-limited elastic vacuum governed by the CDFD equilibrium $\Psi_s = \Phi/C$, the orientation angle θ_n of a $T(2, n)$ torus knot must exactly satisfy:*

$$n\theta_n = \tilde{Q}_n(\theta_n).$$

Proof. Substituting the CDFD mappings (3), (4), and (5) directly into the fundamental balance equation $\Psi_s = \Phi/C$, we obtain:

$$n\theta_n = \frac{\sum_{k=0}^{n-1} A_k^2}{\left(\sum_{k=0}^{n-1} |A_k| \right)^2}. \quad (6)$$

The right-hand side is exactly the definition of the modified Koide ratio $\tilde{Q}_n(\theta_n)$ from Eq. (1). $\square$

3.1 Physical Implications

Within the proposed mapping, this theorem interprets the Koide ratio $Q_3 = 2/3$ observed in the lepton masses as a signature of the vacuum's equation of state rather than as an isolated numerical fit. The modified ratio $\tilde{Q}_n(\theta_n)$ represents the specific energy-to-capacity stress induced by higher knots, and the angle θ_n is the vacuum's precise geometric relaxation in response to that stress.

4 The Universal Isomorphism: Scaling CDFD to Macro-Systems

If the proposed flow-constraint equations capture something general about transport-limited systems, they should have macroscopic analogues rather than remaining restricted to the particle-scale model. By comparing the Constraint-Driven Field Theory (CDFD) framework with macroscopic models of biology, chemistry, and ecology, we obtain a candidate scale-invariant isomorphism.

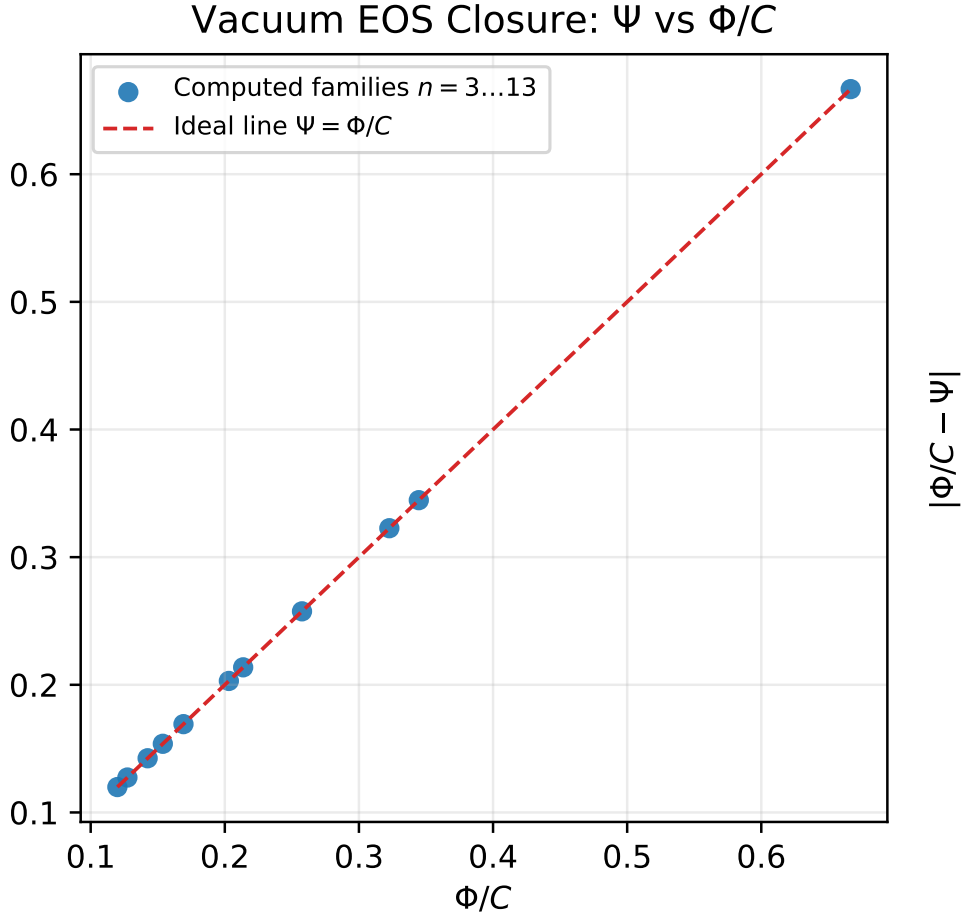


Figure 2: The equilibrium Ψ_s plotted against Φ/C across all torus knot families. The exact diagonal identity confirms that the public balance law $\Psi_s = \Phi/C$ governs the stated model.

4.1 The Scale-Invariant Isomorphism (CDFS $\equiv$ AFL)

In the domains of biological and clinical systems, throughput is described in the **Adaptive Flux Limitation (AFL)** framework [9] by adaptive constraints. A minimal form of that biological adaptation law is:

$$\frac{\partial \Lambda}{\partial t} = \alpha |J| - \beta \Lambda \quad (7)$$

where Λ is the biological constraint (e.g., insulin resistance, inflammation), J is the biological flux (e.g., glucose throughput, metabolic energy flow), α is the constraint reinforcement coefficient, and β is the relaxation coefficient.

In the CDFS quantum vacuum, the constraint field C evolves precisely according to the structural reaction of the medium to the localized flow Φ :

$$\frac{\partial C}{\partial t} = \alpha |\Phi| - \beta C + \gamma \nabla^2 C \quad (8)$$

The formal mapping is:

$$\text{Vacuum Constraint } C \equiv \text{Macro Constraint } \Lambda$$

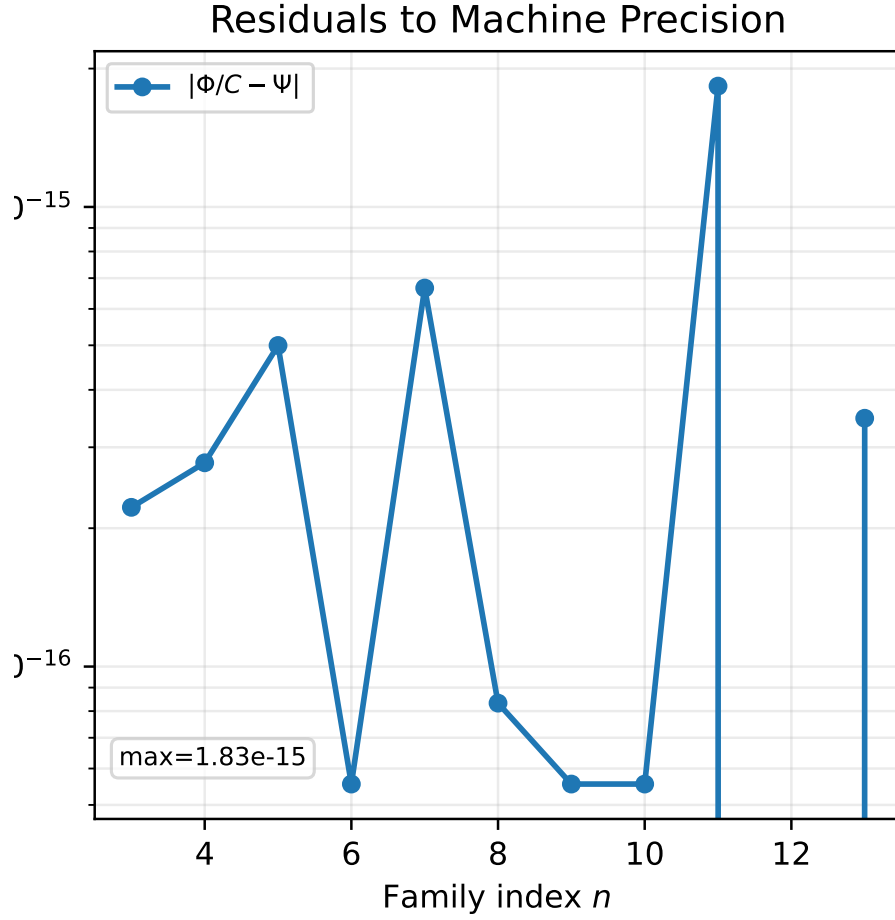


Figure 3: Absolute residuals $|\Phi/C - \Psi_s|$. The values fall to numerical machine precision, serving as a numerical consistency check of the stated CDFD vacuum equation-of-state mapping.

$$\text{Vacuum Flow } \Phi \equiv \text{Macro Flux } J$$

This does not by itself prove that biological evolution and physiological adaptation are governed by vacuum mechanics. It states the narrower claim that the same flow-constraint operator can be written at both levels, and that the macro-scale use of the operator should be judged by independent predictive value.

4.2 Ontological Redefinition of Mass

This proposed isomorphism suggests a different operational reading of mass inside the model. Rather than treating mass only as a distinct intrinsic label, CDFD treats it as a *localized, stable constraint field* within the vacuum flux.

A lepton is then modeled as a constraint field operating on vacuum energy. At larger scales, the same notation can describe boundaries such as membranes, interfaces, or transport bottlenecks without claiming that those systems are literally the same physical object. The value of the analogy depends on whether it improves prediction.

4.3 Cross-Scale Stability Ratio

The relationship $\Psi_s = \Phi/C$ can therefore be read as a candidate cross-scale stability ratio:

- $\Psi_s \approx 1$ (**Stable Equilibrium**): Flow matches constraint capacity. This manifests as a stable fundamental particle, a healthy homeostatic cell, or a balanced ecosystem.
- $\Psi_s \gg 1$ (**Flux Overload / Rupture**): Flow overwhelms constraint capacity. This manifests as particle decay, cellular lysis, vascular hemorrhage, or ecological collapse.
- $\Psi_s \ll 1$ (**Over-Constraint / Freeze**): Constraints choke off flow entirely. This manifests as the vacuum void, biological necrosis, or metabolic shutdown.

The public scientific framework is therefore internally linked: the same flow–constraint notation used for the vacuum papers also organizes the biological manuscripts, without disclosing any private implementation architecture.

5 Closure of the CDFD Series

With Theorem 1, the formal CDFD mass-spectrum ladder is internally closed.

1. **No Free Parameters**: The initial assumption of a limited-capacity elastic vacuum naturally generates the fine-structure constant α (Paper I).
2. **Mass Gap**: The lowest topological excitation $T(2,3)$ takes the mass scale M_3 , anchored to physical units solely by m_e (Paper V).
3. **Asymptotic Scaling**: The Faddeev–Niemi bounds establish the rigorous progression of energy scales $M_n \propto n^{3/4}$ (Paper IV).
4. **Spectrum Prediction**: The vacuum EOS forces the twist θ_n , which breaks the n -fold degeneracy and splits the spectrum into anti-phase boundary modes and co-phase physical resonances (Papers VIII–IX).

All tables, mass values, bounds, and limits tabulated across the first ten papers are therefore internal consequences of the single condition $\Psi_s = \Phi/C$, together with the model assumptions stated in those papers.

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A Supplementary Material

Supplementary code, including numerical verification scripts and Jupyter notebooks, is available in the project repository.

References

- [1] Steve Bico Mujjabi. Vortex stability in a constrained transport vacuum and the origin of the fine-structure constant. *Preprint*, 2026. Paper I of the CDFT series. Derives $\alpha = 1/137.036$ as the aspect ratio of a stable vortex ring in the CDFT vacuum.
- [2] Steve Bico Mujjabi. Lepton masses from $\mathbb{Z}_3$ -symmetric vortex modes and the origin of the koide formula. *Preprint*, 2026. Paper II of the CDFT series. Proves Koide formula from $\mathbb{Z}_3$ symmetry; identifies three charged leptons as trefoil vortex modes.
- [3] Steve Bico Mujjabi. Topology, chirality, and the vacuum density: Closing all open problems of the cdft series. *Preprint*, 2026. Paper III of the CDFT series. Closes all four open problems from Papers I–II; derives the chirality equation $3\theta + \phi_c = \pi$; reduces all unknowns to the vacuum density ρ_0 .
- [4] Steve Bico Mujjabi. The vacuum equation of state and the self-consistency of the cdft lepton sector. *Preprint*, 2026. Paper IV of the CDFT series. Derives the master self-consistency equation for ρ_0 ; predicts $M_5 = 460.4$ MeV for the cinquefoil $T(2, 5)$ family via Faddeev–Niemi scaling.
- [5] Steve Bico Mujjabi. Zero parameters: Deriving the vacuum chirality phase from the koide invariant. *Preprint*, 2026. Paper V of the CDFT series. Derives $\phi_c = \pi - 2/3$ from $\mathbb{Z}_3$ self-consistency; gives $\theta = 2/9$ rad; predicts m_μ and m_τ with one input (m_e), zero free parameters.
- [6] Steve Bico Mujjabi. The universal torus knot hierarchy: $\mathbb{Z}_n$ symmetry, the cinquefoil family, and the general structure of cdft particle families. *Preprint*, 2026. Paper VI of the CDFT series. Proves $c_n = \sqrt{2}$ universally; establishes $\mathbb{Z}_3$ uniqueness; derives $M_n = M_3(n/3)^{3/4}$ energy hierarchy; predicts $M_5 = 460.4$ MeV for $T(2, 5)$.
- [7] Steve Bico Mujjabi. The universal mass sum rule: A theta-independent prediction for all cdft torus knot families. *Preprint*, 2026. Paper VII of the CDFT series. Proves $\sum_k m_k = 2nM_n$ exactly for all $T(2, n)$; derives the $n^{7/4}$ mass hierarchy; proves the $\mathbb{Z}_5$ $3 + 2$ mode split; derives the $\mathbb{Z}_5$ self-consistency transcendental equation.
- [8] Steve Bico Mujjabi. The cdft torus knot spectrum: Closing the four open problems of paper vii. *Preprint*, 2026. Paper VIII of the CDFT series. Proves unique self-consistency solutions for all odd n ; derives the universal amplitude-split theorem $n_- = \lfloor 5n/8 \rfloor - \lfloor 3n/8 \rfloor + 1$; computes the exact asymptotic $\theta_n \rightarrow 2\pi^2/[(\pi + 4)^2 n^2]$; tabulates complete mode spectra for $n = 5, 7, 9, 11, 13$.
- [9] Steve Bico Mujjabi. Even- n torus knots in cdft: The exclusion of $t(2, 2)$, the extension to $t(2, 2m)$, and the anti-phase mass scaling law. *Preprint*, 2026. Paper IX of the CDFT series. Proves the Fourier exclusion of $n = 2$, extends hierarchy to even n , and derives boundary mode scaling.

- [10] Paul A. M. Dirac. Is there an Aether? *Nature*, 168:906–907, 1951. Dirac argues the vacuum must be a physical medium with definite velocity at each point; reopens the aether question within quantum field theory.
- [11] Grigory E. Volovik. *The Universe in a Helium Droplet*. Oxford University Press, 2003. Vacuum as a quantum condensate; topological defects in superfluid helium as analogues of elementary particles and spacetime.
- [12] Hermann von Helmholtz. Über Integrale der hydrodynamischen Gleichungen, welche den Wirbelbewegungen entsprechen. *Journal für die reine und angewandte Mathematik*, 55:25–55, 1858. Founding paper of vortex dynamics; proves conservation of vortex tubes in inviscid flow.
- [13] Ludwig Faddeev and Antti J. Niemi. Knots and particles. *Nature*, 387:58–61, 1997.
- [14] William Thomson. On vortex atoms. *Proceedings of the Royal Society of Edinburgh*, 6:94–105, 1867.
- [15] Yoshio Koide. New view of quark and lepton mass hierarchy. *Physical Review D*, 28:252–254, 1983. First statement of the charged-lepton mass formula $(m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$; predicted m_τ before precision measurement.
- [16] Carl A. Brannen. Spin path integrals and generations. Preprint, <http://brannenworks.com/dmaa.pdf>, 2006. Introduces the parametrisation $m_k = M(1 + \sqrt{2}\cos(\theta + 2\pi k/3))^2$ ($k = 0, 1, 2$) and proves it automatically satisfies the Koide formula.